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Testing for spatial market integration: evidence for Colombia using a pairwise approach

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Abstract

We examine the extent of spatial market integration in Colombia using consumer price index data for 153 consumer goods in 13 cities. An econometric analysis of the time-series properties of all the possible city price differentials reveals that market integration tends to occur more frequently in unprocessed food products, as opposed to processed foods, other traded and nontraded products. The results also support the view that, except for nontraded products, the speed at which prices adjust to the long-run equilibrium is slower for cities that are farther apart.

JEL classifications: C33, E31, Q11

Keywords: Law of one price; Pairwise; Distance; Colombia

1. Introduction

Ever since the writings of Cournot (1838) and Marshall (1920), the question of whether the price of a good in spatially separated locations is the same, after allowance is made for the quality differences, transportation costs, and other barriers to trade, has received a great deal of attention. This is because spatial integration helps define the extent of the market, that is, the area within which the interaction of producers and consumers determines the price of a product; see, e.g., Stigler (1949) for an early reference, or Stigler and Sherwin (1985) for an exposition with several empirical illustrations. In a well-functioning market, competition guarantees that the effect of a shock to the price of a product in one location propagates to other locations within that market as well, so that producers and consumers are able to exploit arbitrage opportunities. Thus, for example, spatial integration permits distant places to absorb excess local supply, preventing excessive price falls that would otherwise hurt the profitability of local producers; see Barrett (2008). As indicated by Fackler and Goodwin (2001), spatial market integration has often been assessed through tests of the validity of the law of one price (LOP), either by examining whether the prices of identical products traded in different locations are the same once they are converted to a common currency, as in the case of the absolute version of the law, or by testing whether price discrepancies are stationary, as in the case of the relative version of the law; see, e.g., Froot and Rogoff (1995) and Sarno and Taylor (2002).

The objective of this article is to investigate the extent of spatial market integration in Colombia. To do this, we use disaggregated consumer price index (CPI) data for different cities to assess the validity of the LOP in its relative version. Our empirical analysis starts off with a time-series approach, but in a way that subsequently also employs information from a cross-sectional dimension. More specifically, the empirical modeling strategy followed in the article involves two main stages. First, we adapt the Pesaran (2007) pairwise approach to test for output and growth convergence to the analysis of spatial market integration. Here, the idea is that given a sample of N prices, unit root tests are conducted on all possible $(N(N-1)/2)$ price differentials, and then the total fraction of rejections is calculated. Second, we examine the determinants of the speed of adjustment toward long-run equilibrium. This considers the role played by the distance between cities insofar as one would expect that the speed of adjustment toward long-run equilibrium is fastest between contiguous as opposed to more distant or noncontiguous cities. In doing this, the role played by the relative size between city pairs is also explored.

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The modeling strategy adopted in this article differs from a traditional approach that involves selecting the price of a product in a base city, measuring the remaining $(N-1)$ price series against that base price, and applying panel unit root tests to the resulting relative prices. Under the traditional framework, however, outcomes may turn out to be highly sensitive to the choice of the base price. In contrast, the pairwise approach offers the advantage that, by incorporating all possible bivariate relationships, it does not involve what can be a problematic choice of a single reference city in the computation of price differentials.

The Pesaran pairwise approach has been recently applied by Yazgan and Yilmazkuday (2011) to the analysis of price-level convergence in the United States using microprice data, and by Yilmazkuday (2013) to investigate inflation rates convergence in Turkey during preinflation-targeting and inflation-targeting periods, where inflation rates are computed using disaggregated-level CPI data.[1] Thus, further research on this important topic for a developing economy such as Colombia appears fruitful, as it would enable us to establish similarities and differences among the three countries.

In addition to this, a distinguishing feature of our article is that we also extend the Pesaran pairwise analysis in an important direction, by formally testing the hypothesis of whether the speed at which intercity price differentials adjust to shocks (and other innovations) is positively related to the geographic separation between cities. This is an aspect that was not considered in the works of Yazgan and Yilmazkuday (2011) and Yilmazkuday (2013). Here, a salient feature of our article is that we employ two variables to measure the geographic separation between cities, namely, the optimal distance (in kilometers) that separates one city from another as one moves along the road network, and the optimal travel duration time (in minutes) that is required to cover the journey. Unlike more traditional measures (e.g., greater circle distance), the use of distance and travel duration time offers the advantage of taking into consideration the complexities of the existing road network of the country, and the diverse topographical conditions of the terrain. In short, we believe that optimal distance and optimal travel duration time can be thought of as good proxies of transportation and transaction costs.

The article proceeds as follows: Section 2 presents a brief review of the literature. Section 3 summarizes the econometric modeling strategy. Section 4 describes the data employed in the empirical modeling exercise. Section 5 presents the results of the pairwise analysis of market integration. Section 6 analyses the determinants of the speed of adjustment of stationary price differentials. Section 7 offers some concluding remarks.

2. A brief review of the literature

Although a large number of authors have empirically tested whether markets are spatially integrated, Fackler and Goodwin (2001), for instance, survey more than 60 studies with a focus on agricultural products, not very many of them have explicitly considered the role of distance as a factor that helps to explain the speed of convergence of price discrepancies. Engel and Rogers (1996), using CPI data from 23 North American cities (14 in the United States and 9 in Canada) for 14 good categories sampled from June 1978 to December 1994, find that the distance between cities and the existence of a border between the United States and Canada are important factors determining the variability of relative prices.

A more disaggregated analysis for the United States is performed by Parsley and Wei (1996) using a database of prices assembled from publications of the American Chamber of Commerce Researchers Association (ACCRA). The database comprises 51 traded and nontraded goods and services in 48 cities, over the period January 1975–April 1992. Parsley and Wei estimate an augmented version of the Levin et al.'s (2002) panel unit root regression that also includes distance as an additional explanatory variable, and find that the rates at which prices converge are slower for cities farther apart. However, it was already indicated that an important shortcoming of panel unit root tests is that the outcome may be sensitive to the selection of the benchmark price. In addition to this, Parsley and Wei ignore potential problems that may be caused by the inclusion of variables such as distance and others that involve nonlinear transformations in the panel unit root test regressions; indeed, the statistical properties of the panel unit root tests in the presence of additional regressors (other than intercept and linear trend terms) are far from fully worked out. In subsequent work, Parsley and Wei (2001) use a three-dimensional panel data set with the prices of 27 traded goods, over the period 1976Q1 through 1997Q4, across 96 cities in the United States and Japan, and find that the distributions of relative prices within these countries are markedly less volatile (and on average close to zero) than the corresponding distribution between them; see also Baba (2007) for an analysis using Japanese and Korean city prices.

The efforts undertaken by the European Union (EU) to integrate national markets, through the adoption of the Single Market Programme, have also provided a fertile ground to test the validity of the LOP, both in its absolute and relative versions. For example, Fousekis (2007) applies a general multivariate cointegration approach to monthly price series of pork and poultry from 14 EU countries, over the period January 1995 to June 2006, and finds that prices in these markets are far from being uniform. Emmanouilides and Fousekis (2012) investigate the validity of the LOP in four major EU pork markets, using weekly data from 1991:w1 to 2008:w52. They find that markets are well integrated and that price deviations from the steady state are corrected in a nonlinear fashion. More recently, Emmanouilides and Fousekis (2015) use weekly price data between 1991:w1 and 2012:w52 to investigate the validity of the

[1] In related literature, authors such as Goletti and Babu (1994), Asche et al. (2012), and Emmanouilides and Fousekis (2012) have examined pairwise price relationships, although not within the Pesaran (2007) framework.

LOP in five major EU broiler markets. After allowing for the possibility of nonlinear adjustment and structural breaks, they find that the LOP holds mostly in its weak version; this implies that although price shocks are fully transmitted across space, price differentials are likely to be present even in the long run.

Yazgan and Yilmazkuday (2011) apply the Pesaran pairwise approach to ACCRA price data over the period 1990Q1–2007Q4, and find not only ample evidence for price-level convergence, but also that the estimated convergence rates have been quicker than previously found in the literature. Yilmazkuday (2013) compares convergence properties of regional inflation rates in Turkey using a disaggregated level CPI data (covering the 10 main expenditure-based groups) before and after the implementation of the inflation targeting regime in January 2002. The results indicate that before inflation targeting, regional inflation rates for CPI groups with more nontraded (traded) products converged to each other to a lesser (greater) extent than during the period of inflation targeting.

In the case of Colombia, Ramírez (1999) provides an initial examination of the idea that the development of the transportation infrastructure lowers the cost of freight and reduces the variability of prices across regions. Using annual price data series for eight agricultural commodities in 12 cities over the period 1928–1990, this author finds that price dispersion across cities declined sharply following the development of transport infrastructure in the 1930s, but that no major further declines occurred after this decade. Barón (2004) applies unit root tests to aggregate consumer price indices of food and housing in seven Colombian cities, and finds evidence of stationarity for the relative prices of food but not for housing; this can be viewed as suggesting that market integration holds for traded goods but not for nontraded ones. Iregui and Otero (2011) apply the Hadri (2000) panel stationarity test to CPI data (January 1999–December 2007) for 54 food products in 13 cities, and find that market integration is favored when cities have similar population and economic characteristics. In addition to this, the speed of adjustment of prices, measured by the half-lives that result from fitting vector autoregressive (VAR) models, and calculating the associated generalized impulse responses, appears faster when food products are perishable.

The role of distance in Colombia is explicitly considered by Iregui and Otero (2013), who use weekly wholesale price data (4 January 2002–18 March 2011) for 18 agricultural products traded in markets scattered around the country, as obtained from the Sistema de Información Agropecuaria assembled by the Corporación Colombia Internacional (CCI). A highly-dimensional VAR model analysis of the half-lives associated with shocks to the price series in pairs of markets suggests that for most products, the distance between markets exerts a positive influence.

An important distinguishing feature of this article is that the spectrum of products that we study is much wider than that considered in previous research. Indeed, the range of products under consideration comprises unprocessed food, processed food, other traded, and nontraded. Here, the inclusion of the nontraded category can be thought of as serving the purpose of being a control group.

3. Econometric modeling exercise

The econometric modeling strategy followed in this article is based on the Pesaran (2007) pairwise approach to testing for output and growth convergence. Let us denote $p_{i,k,t}$ as the price of good k in city i at time t, and define for a given city pair (i, j) and a given good k at time t, the good-level city price differential (or gap) as $g_{ij,k,t} = p_{i,k,t} - p_{j,k,t}$, where $i = 1, ..., N-1$, $j = i+1, ..., N$, and N is the total number of cities. The idea underlying the pairwise approach is the examination of the time-series properties (that is, the order of integration) of all $(N(N-1)/2)$ possible price differentials. The important point to notice here is that such an approach does not suffer from the potential pitfalls that may be involved when selecting the city price benchmark with respect to which all other prices are going to be measured. In other words, it may well be the case that the price of a good in city i is found as stationary when measured against the price observed in city j, but not when measured against the price in another city.

To examine the time-series properties of $g_{ij,k,t}$, we employ the Augmented Dickey-Fuller (ADF) and the $\text{ADF}_{\max}$ unit root tests developed by Dickey and Fuller (1979) and Leybourne (1995), respectively. The ADF test is the well-known regression-based procedure which, in its simplest form, tests the unit-root null hypothesis of nonstationarity, i.e., that the first-order autoregressive coefficient in a first-order autoregressive equation is equal to one, against the alternative of stationarity, i.e., that it is less than one. In turn, the $\text{ADF}_{\max}$ test is based on the maximum value of the standard ADF tests when applied to both forward and reversed realizations of the data. This basic setup can be suitably extended, for instance, by including an intercept term in the unit-root test regression, where rejection of the null implies that $p_{i,k,t}$ and $p_{j,k,t}$ move together in the long run, but not necessarily in a way such that they are same. Furthermore, it is important to notice that by examining $g_{ij,k,t} = p_{i,k,t} - p_{j,k,t}$, we are implicitly imposing the LOP that implies, from an economic point of view, that the goods are perfect substitutes. An alternative view would be to examine if the relationship between $p_{i,k,t}$ and $p_{j,k,t}$ is such that the slope coefficient is different from unity, in which case the goods would be imperfect substitutes. This relationship was first introduced in a general form by Richardson (1978); see also Asche et al. (2012).

Within this framework, consider the ADF ($\text{ADF}_{\max}$) test applied to $g_{ij,k,t}$, and define the indicator function $Z_{ij,k} = 1$ if the unit-root null is rejected, at a significance level α. Next, Pesaran (2007) studies the fraction of the $(N(N-1)/2)$ gaps for which the unit root hypothesis is rejected, that is:

$$\overline{Z}_k = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} Z_{ij,k}, \tag{1}$$

and shows that under the null hypothesis of divergence (i.e., nonstationarity), the expected value of $\overline{Z}_k$ must be equal to the nominal size of the underlying unit root test statistic, α. By contrast, under the null of convergence, the proportion of rejections $\overline{Z}_k$ is expected to be high and approach 100% as $T \to \infty$.

While computation of the fraction of rejections $\overline{Z}_k$ may in itself constitute a parameter of interest, as in, e.g., Pesaran (2007) and Yazgan and Yilmazkuday (2011), we develop the pairwise approach further by considering the factors that might help to explain the speed at which relative prices adjust to shocks. To do so, we focus on the intercity price differentials $g_{ij,k,t}$ for which the unit root null is rejected using the ADF test at the 10% significance level. For these stationary cases, we compute the approximated half-life of a shock (in months) using the well-known formula $-\ln(2)/\ln(1+\hat{\delta})$, where $\hat{\delta}$ indicates the autoregressive coefficient in the corresponding ADF test regression; see, e.g., Goldberg and Verboven (2005). The resulting half-life is denoted by $hl_{ij,k}$ and, in a subsequent stage of the analysis, a regression model is postulated to investigate the factors that may help to explain its variability.

4. Data

The database, obtained from the Departamento Administrativo Nacional de Estadística (DANE), consists of seasonally unadjusted monthly observations on consumer price indices for 153 products in the 13 main metropolitan areas of the country, namely, Bogotá, Medellín, Cali, Barranquilla, Cartagena, Cúcuta, Bucaramanga, Pereira, Villavicencio, Pasto, Montería, Manizales, and Neiva (where the cities have been listed in terms of their population, from largest to smallest). For this study, and in order to facilitate the presentation of our results, we follow the classification of products used by the Banco de la República (the country's central bank) in its quarterly inflation report; see, e.g., Banco de la República (2013). Thus, we consider two broadly defined product categories, namely, traded and nontraded; the former consists of 125 products, while the latter 28. Within the traded category, we make a further distinction between unprocessed food (17), processed food (37), and other traded products (71), where the factor underlying the distinction among the three subgroups is the degree of perishability of the good. The first column in Tables 1 and 2 provides the detailed list of all products included in this study.[2] The sample period spans from January 1999 to February 2013, for a total of 170 time observations, and all price indices are considered after applying the logarithmic transformation.

[2] The products under consideration account for more than 90% of the overall CPI. The remaining share corresponds to products whose prices are regulated (such as utilities, petrol, and bus fares, among others), and therefore exhibit little variation over time and/or across metropolitan areas.

Table 1
Proportion of stationary price differentials

Product	ADF	$\text{ADF}_{\max}$
Unprocessed food	0.711	0.795
Bananas	0.410	0.513
Blackberries	0.744	0.821
Carrots	0.756	0.846
Cassava	0.718	0.859
Eggs (all sizes)	0.551	0.603
Fresh vegetables	0.679	0.731
Kidney beans (fresh, dried)	0.667	0.833
Onion (all varieties)	0.795	0.833
Oranges	0.962	1.000
Other dried vegetables (lentils, chickpea)	0.179	0.295
Other fresh fruits	0.590	0.692
Other roots	0.821	0.949
Peas (fresh, dried)	0.731	0.897
Plantain	0.808	0.808
Potato (all varieties)	0.987	0.974
Tomato	0.910	0.987
Tree tomato	0.782	0.872
Processed food	0.353	0.329
Beef (all cuts)	0.308	0.423
Bread (all varieties)	0.218	0.269
Burger in bun—eat in	0.244	0.167
Canned vegetables (kidney beans, peas)	0.526	0.090
Canteen meals	0.308	0.244
Cereals (all varieties)	0.333	0.218
Cheese (all types)	0.551	0.449
Chicken (all cuts)	0.346	0.269
Chocolate	0.244	0.256
Coffee	0.410	0.449
Cold fast food	0.179	0.077
Fats	0.333	0.397
Fish (all types)	0.436	0.500
Flour (all varieties)	0.333	0.372
Hot fast food	0.218	0.154
Juices	0.423	0.462
Milk (all types)	0.436	0.372
Oils (soya, maize, sunflower)	0.628	0.487
Other bakery products	0.256	0.333
Other cereals	0.308	0.244
Other condiments	0.385	0.115
Other dairy products	0.397	0.513
Other fish products	0.462	0.244
Other groceries (crisps, jam)	0.154	0.231
Other meats	0.346	0.295
Other nonalcoholic beverages	0.256	0.269
Panela (sugar cane by-product)	0.321	0.256
Pasta (all varieties)	0.372	0.526
Pork (all cuts)	0.321	0.359
Restaurant meals	0.333	0.167
Rice (all varieties)	0.628	0.667
Salt	0.436	0.526
Soft drinks	0.308	0.333
Soups and creams	0.282	0.346
Sugar	0.397	0.487
Various jams	0.333	0.282
Various sauces	0.308	0.321
Other traded	0.248	0.160
Aguardiente (spirit)	0.192	0.077
Air fares	0.872	0.038

(*Continued*)

Table 1
Continued

Product	ADF	ADF_{max}
Audio systems and recorders	0.154	0.167
Babies clothing (underwear)	0.026	0.051
Bed linen	0.231	0.179
Beer	0.167	0.179
Blankets and bedspreads	0.218	0.103
Books	0.256	0.128
Boy's trousers	0.192	0.077
Car batteries	0.128	0.167
Car service (motor oil)	0.192	0.103
Car tyres	0.192	0.154
Compact disks	0.295	0.308
Children's clothing (trousers)	0.167	0.064
Children's clothing (various shirts)	0.256	0.090
Children's footwear	0.051	0.000
Cigarettes	0.167	0.205
Cleaning products and disinfectants	0.538	0.423
Clocks and watches	0.269	0.179
Cooker (all types)	0.244	0.128
Corporal hygiene products	0.385	0.333
Crockery set	0.128	0.115
Curtains	0.179	0.038
Cutlery set	0.167	0.115
Dishwasher	0.179	0.231
Equipment for house	0.128	0.090
Facial care (various cosmetics)	0.333	0.205
Fridge/freezer	0.282	0.128
Hair care products	0.359	0.192
Insecticides	0.154	0.205
Jewellery (gold and silver)	0.205	0.090
Kitchen equipment	0.154	0.179
Mattresses and pillows	0.333	0.205
Medicines	0.192	0.026
Men's clothing (trousers and jeans)	0.218	0.141
Men's clothing (underpants, t-shirts, socks)	0.256	0.103
Men's clothing (various shirts)	0.154	0.051
Men's footwear	0.115	0.051
Nappies and others	0.321	0.321
Newspapers	0.295	0.103
Nondurable kitchen goods	0.269	0.321
Notebooks	0.551	0.359
Oral hygiene products	0.231	0.218
Orthopedic equipment and others	0.154	0.064
Other alcoholic drinks	0.333	0.244
Other audiovisual equipment	0.179	0.064
Other children's clothing	0.359	0.103
Other household electric appliances	0.115	0.051
Other medicines and contraceptives	0.603	0.526
Other men's clothing	0.218	0.051
Other motoring expenditure	0.192	0.218
Other personal effects	0.064	0.051
Other personal grooming products	0.346	0.218
Other recreational items	0.077	0.000
Other school expenses	0.333	0.256
Other school materials	0.244	0.090
Other telephone services	0.244	0.141
Other women's clothing	0.231	0.026
Periodicals	0.372	0.449
Soaps	0.282	0.359
Sport shoes	0.128	0.103
Sports equipment (e.g., bicycles)	0.295	0.103

(*Continued*)

Table 1
Continued

Product	ADF	ADF_{max}
Televisions	0.282	0.269
Textbooks	0.372	0.269
Towels, tablecloths, and furniture covers	0.256	0.231
Washing powder, bleach, and fabric conditioners	0.410	0.551
Waxes (solid and liquid)	0.372	0.167
Women's clothing (blouses)	0.115	0.038
Women's clothing (trousers and jeans)	0.154	0.038
Women's clothing (underwear)	0.231	0.026
Women's footwear	0.256	0.026
Nontraded	0.270	0.151
Bedroom furniture	0.167	0.077
Car park charges	0.218	0.013
Dining room furniture	0.218	0.115
Dry cleaning	0.167	0.077
Enrolment fees	0.167	0.218
Enrolment fees (higher education)	0.436	0.205
Estimated rents	0.026	0.000
Gaming (e.g., lottery tickets)	0.346	0.179
Hairdressing fees	0.256	0.077
Hospital and ambulance charges	0.269	0.115
Lab exams	0.154	0.103
Living room furniture	0.179	0.128
Other education expenses	0.526	0.244
Other household furniture	0.269	0.218
Other personal grooming services	0.269	0.077
Postal charges	0.205	0.244
Prepaid medicine and health insurance	0.513	0.115
Private general medical and dental examinations	0.103	0.051
Private specialized medical examination	0.308	0.090
Recreational services	0.372	0.346
Rents	0.038	0.026
School fees	0.321	0.333
Tailoring and hiring of clothing	0.167	0.038
Television services	0.513	0.115
Tourism	0.244	0.192
Various bank charges	0.615	0.667
Vehicle maintenance and repairs	0.218	0.064
X-rays, ultrasound scans, and electrocardiograms	0.282	0.103

Note: The underlying unit-root test regressions include an intercept, and the number of lags is selected using the Akaike information criterion with $p_{max} = 3$ lags. Both the ADF and ADF_{max} tests are performed at the 10% significance level. For each category, the proportion of stationary price differentials corresponds to the simple average of the proportion of stationary price differentials of the corresponding products.

5. Pairwise tests of market integration

Table 1 reports the fraction of rejections of both the ADF and ADF_{max} unit root tests for the four product categories under consideration, based on all possible $(13 \times 12)/2 = 78$ price differentials. The tests are performed at the 5% and 10% significance levels (although only the latter are reported to save space), include an intercept as deterministic component (so that rejection of the null implies that the price differential is level stationary), and the optimal lag length is determined using the

Table 2
Estimates of the half-life of a shock

Product	*hl*	Duration in months			
		0–3	3–6	6–12	12– +
Unprocessed food	4.3		♦		
Bananas	4.8		♦		
Blackberries	2.9	♦			
Carrots	3.2		♦		
Cassava	7.8			♦	
Eggs (all sizes)	4.5		♦		
Fresh vegetables	4.0		♦		
Kidney beans (fresh, dried)	6.4			♦	
Onion (all varieties)	3.9		♦		
Oranges	3.3		♦		
Other dried vegetables (lentils, chickpea)	6.2			♦	
Other fresh fruits	4.4		♦		
Other roots	7.0			♦	
Peas (fresh, dried)	3.0	♦			
Plantain	4.0		♦		
Potato (all varieties)	3.0		♦		
Tomato	1.6	♦			
Tree tomato	3.7		♦		
Processed food	6.9			♦	
Beef (all cuts)	7.1			♦	
Bread (all varieties)	7.1			♦	
Burger in bun—eat in	6.9			♦	
Canned vegetables (kidney beans, peas)	8.1			♦	
Canteen meals	7.5			♦	
Cereals (all varieties)	5.7		♦		
Cheese (all types)	4.2		♦		
Chicken (all cuts)	5.8		♦		
Chocolate	5.3		♦		
Coffee	5.1		♦		
Cold fast food	7.8			♦	
Fats	5.9		♦		
Fish (all types)	5.1		♦		
Flour (all varieties)	8.9			♦	
Hot fast food	8.0			♦	
Juices	9.0			♦	
Milk (all types)	8.2			♦	
Oils (soya, maize, sunflower)	6.1			♦	
Other bakery products	6.0		♦		
Other cereals	7.7			♦	
Other condiments	17.4				♦
Other dairy products	5.6		♦		
Other fish products	6.8			♦	
Other groceries (crisps, jam)	6.9			♦	
Other meats	6.1			♦	
Other nonalcoholic beverages	7.4			♦	
Panela (sugar cane by-product)	8.8			♦	
Pasta (all varieties)	6.2			♦	
Pork (all cuts)	6.5			♦	
Restaurant meals	9.4			♦	
Rice (all varieties)	4.0		♦		
Salt	5.8		♦		
Soft drinks	5.5		♦		
Soups and creams	5.9		♦		
Sugar	5.3		♦		
Various jams	5.4		♦		

(*Continued*)

Table 2
Continued

Product	*hl*	Duration in months			
		0–3	3–6	6–12	12– +
Various sauces	6.2			♦	
Other traded	10.5			♦	
Aguardiente (spirit)	9.4			♦	
Air fares	9.0			♦	
Audio systems and recorders	9.3			♦	
Babies clothing (underwear)	7.6			♦	
Bed linen	9.1			♦	
Beer	7.3			♦	
Blankets and bedspreads	15.8				♦
Books	9.6			♦	
Boy's trousers	12.5				♦
Car batteries	6.9			♦	
Car service (motor oil)	10.2			♦	
Car tyres	8.2			♦	
Compact disks	5.9		♦		
Children's clothing (trousers)	8.6			♦	
Children's clothing (various shirts)	16.0				♦
Children's footwear	17.1				♦
Cigarettes	7.6			♦	
Cleaning products and disinfectants	7.1			♦	
Clocks and watches	8.8			♦	
Cooker (all types)	9.1			♦	
Corporal hygiene products	7.3			♦	
Crockery set	8.3			♦	
Curtains	19.3				♦
Cutlery set	8.8			♦	
Dishwasher	9.5			♦	
Equipment for house	12.4				♦
Facial care (various cosmetics)	8.7			♦	
Fridge/freezer	8.3			♦	
Hair care products	7.1			♦	
Insecticides	8.1			♦	
Jewellery (gold and silver)	9.1			♦	
Kitchen equipment	6.6			♦	
Mattresses and pillows	8.5			♦	
Medicines	21.1				♦
Men's clothing (trousers and jeans)	11.6			♦	
Men's clothing (underpants, t-shirts, socks)	12.5				♦
Men's clothing (various shirts)	13.7				♦
Men's footwear	21.1				♦
Nappies and others	6.2			♦	
Newspapers	11.1			♦	
Nondurable kitchen goods	6.7			♦	
Notebooks	6.9			♦	
Oral hygiene products	9.8			♦	
Orthopedic equipment and others	11.7			♦	
Other alcoholic drinks	8.8			♦	
Other audiovisual equipment	8.0			♦	
Other children's clothing	16.2				♦
Other household electric appliances	9.1			♦	
Other medicines and contraceptives	7.3			♦	
Other men's clothing	12.6				♦
Other motoring expenditure	6.6			♦	
Other personal effects	21.0				♦
Other personal grooming products	9.1			♦	
Other recreational items	3.9		♦		
Other school expenses	10.3			♦	

(*Continued*)

Table 2
Continued

Product	*hl*	Duration in months			
		0–3	3–6	6–12	12– +
Other school materials	12.7				♦
Other telephone services	6.0			♦	
Other women's clothing	14.4				♦
Periodicals	7.6			♦	
Soaps	5.9		♦		
Sport shoes	10.6			♦	
Sports equipment (e.g., bicycles)	11.5			♦	
Televisions	8.1			♦	
Textbooks	7.5			♦	
Towels, tablecloths' and furniture covers	9.7			♦	
Washing powder, bleach' and fabric conditioners	5.1		♦		
Waxes (solid and liquid)	8.2			♦	
Women's clothing (blouses)	12.6				♦
Women's clothing (trousers and jeans)	20.4				♦
Women's clothing (underwear)	22.3				♦
Women's footwear	21.6				♦
Nontraded	10.6			♦	
Bedroom furniture	8.1			♦	
Car park charges	15.1				♦
Dining room furniture	10.4			♦	
Dry cleaning	13.6				♦
Enrolment fees	4.7		♦		
Enrolment fees (higher education)	8.3			♦	
Estimated rents	19.4				♦
Gaming (e.g., lottery tickets)	5.3		♦		
Hairdressing fees	13.4				♦
Hospital and ambulance charges	11.7			♦	
Lab exams	10.8			♦	
Living room furniture	8.8			♦	
Other education expenses	9.5			♦	
Other household furniture	9.7			♦	
Other personal grooming services	13.1				♦
Postal charges	4.5		♦		
Prepaid medicine and health insurance	9.2			♦	
Private general medical and dental examinations	23.8				♦
Private specialized medical examination	16.2				♦
Recreational services	7.7			♦	
Rents	13.6				♦
School fees	4.8		♦		
Tailoring and hiring of clothing	11.4			♦	
Television services	10.9			♦	
Tourism	7.8			♦	
Various bank charges	3.7		♦		
Vehicle maintenance and repairs	11.7			♦	
X-rays, ultrasound scans, and electrocardiograms	10.1			♦	

Note: The half-life of each category is measured in months, and corresponds to the simple average of the half-lives of the individual products.

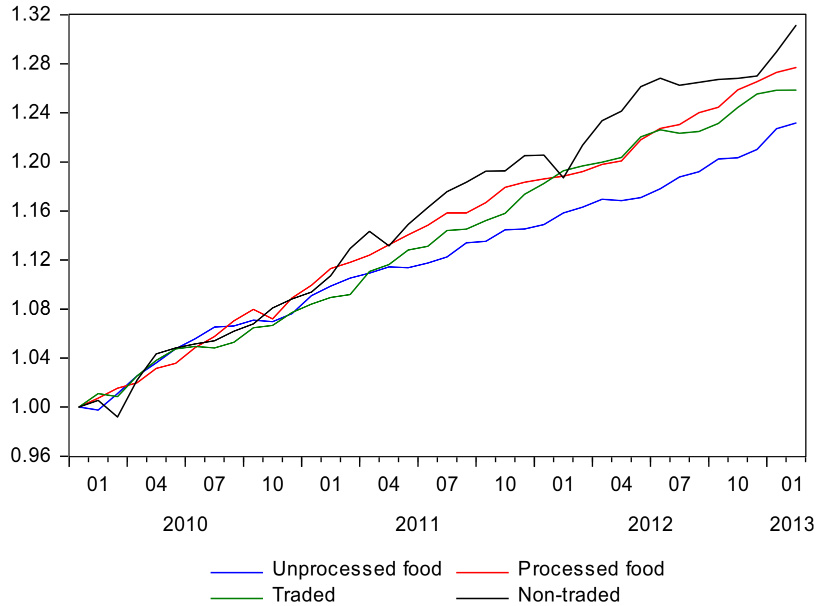
Akaike information criterion (AIC), with $p_{\max} = 3$ lags. Considering first the results of the ADF test, it is interesting to see that the fraction of rejections varies depending on the type of product, being largest for unprocessed foods (71%), and followed by that of processed foods (35%). Smaller fractions of stationary price differentials are obtained for nontraded and other traded goods with 27% and 25%, respectively. Qualitatively similar percentages of rejection are obtained when inference is based on the $\text{ADF}_{\max}$ test, namely, 80%, 33%, 16%, and 15% for unprocessed foods, processed foods, other traded goods, and nontraded goods, respectively.

Our findings thus suggest that storable products, which normally have lower price volatility, exhibit limited market integration when compared to food products. One possible explanation is that unprocessed foods are more homogeneous perishable products, and as a result of this arbitrage opportunities are more fully exploited through corrective movements by buyers and/or sellers. Indeed, the strongest evidence of spatial market integration is observed in foods such as potato (all varieties), oranges, tomato, and other roots, where in all cases, the percentages of stationary price differentials are above 90%.

At this stage of the analysis, and despite the fact that Yazgan and Yilmazkuday (2011) employ microprice data while we use CPI-based data, it is worth comparing our aggregate results with those obtained by these authors for the United States (also assuming that the product categories in the two studies consist of similar goods). In particular, these authors find that the percentages of rejection for nonperishable and nontraded good categories are equal to 60% and 44%, respectively, which are higher than our estimates for processed and nontraded goods reported above (where the comparison is based on the results obtained for the ADF test at the 10% level of significance, using the AIC to select the optimal numbers of lags). By contrast, the percentage of rejection obtained by Yazgan and Yilmazkuday for perishable goods (i.e., 58%) is lower than our estimate for unprocessed food goods.

The results reported by Yilmazkuday (2013) for Turkey, which are also based on disaggregated CPI data, indicate that the point percentage of rejections is much greater than those reported above for Colombia. For example, in the case of some nontraded goods, the portion of inflation differentials that are stationary reaches 100%. However, we argue that the greater percentages of rejection that are obtained in Turkey can be explained by the fact that Yilmazkuday is interested in the convergence of inflation rates, while we are interested in price levels.

Next, we examine whether the speed at which relative prices adjust to exogenous shocks is also related to the type of good. To do this, for each product, k, we consider only the intercity price differentials that are stationary, based on the ADF test at the 10% significance level, and for these cases, we compute the approximate half-life of a shock, which we denote as $hl_{ij,k}$. Then, the resulting half-lives are averaged by product and then by product category; see Table 2.



Note: The half-lives of the shocks have been normalized so that they are equal to 1 in 2010m01.

Fig. 1. Recursive estimates of the half-life of a shock. [Color figure can be viewed at wileyonlinelibrary.com]

Assuming that the half-life of a shock is inversely related to the speed of adjustment of the corresponding relative price, our findings suggest that the speed of adjustment is quickest for unprocessed and processed food products, where the average half-lives are approximately 4.3 and 6.9 months, respectively. The products for which adjustment is the quickest (i.e., less than 3 months) are blackberries, peas (fresh, dried), and tomato. It is interesting to note that, despite differences in methodology and study period, our findings for food products corroborate those obtained by Iregui and Otero (2011). Regarding traded and nontraded products, the resulting average half-lives reported in Table 2 are very similar (about eight months). We argue that the low speed of price adjustment for traded goods in Colombia may be due to the fact that the products in this category are mainly manufactures, whose prices tend to have more of a fixed-price characteristic because of market concentration in the industrial sector. By contrast, Ardeni and Freebairn (2002) point out that agricultural product prices tend to be more of the flex-price type because of the presence of (almost) perfect competition, the influence of seasonal and climatic fluctuations, the fact that most agricultural activities are greatly dispersed on the territory, and the restrictions imposed by limited technological progress in the sector. This result is in sharp contrast to Yazgan and Yilmazkuday (2011), where the convergence rate for nontraded goods is estimated at about twice the rates obtained for perishable and nonperishable goods.

In addition to the results over the whole sample period, we also assess the time-varying nature of the speed of adjustment as successively larger subsamples are employed for estimation. Thus, Fig. 1 summarizes the results when the half-lives of the shocks are estimated recursively, starting with the sample subperiod 1999m1 to 2010m1, then 1999m1 to 2010m2, and so on until 1999m1 to 2013m2 (for each product category, the resulting half-lives are scaled by dividing them by the corresponding value in 2010m1). As can be seen from the plot, the half-lives exhibit an upward movement indicating slower speed of adjustment as time passes. This upward trend is particularly noticeable for processed food, other traded and nontraded products, and to a lesser extent for unprocessed food products. Results at the product level (not reported here for brevity) allow us to detect a similar upward trend in the overwhelming majority of the cases, with the exceptions occurring in the unprocessed food category in items such as blackberries, cassava, plantain, and potato (all varieties). This finding suggests an economy where stagnating transport linkages fail to transmit local shocks to other markets.

6. Determinants of the speed of adjustment

The analysis that follows is based on the underlying idea that price adjustments ought to take longer for cities that are farther apart. Often, the variable that has been employed in the literature to measure this effect is the (greater circle) distance between city pairs i and j; see, e.g., Engel and Rogers (1996) and Parsley and Wei (1996). An important issue here is in recognizing that this measure does not take into account the complexities of the road network of a country, nor the fact that covering a given distance may take different times depending

Table 3
Determinants of the half-life of intercity price differentials

	Food				Other traded		Nontraded	
	Unprocessed		Processed					
Intercept	0.368	0.418	1.085	1.169	2.429	2.477	3.157	3.177
	(0.176)	(0.164)	(0.174)	(0.163)	(0.237)	(0.231)	(0.290)	(0.266)
ln $durt_{ij}$	0.107		0.121		0.059		−0.037	
	(0.026)		(0.024)		(0.023)		(0.046)	
ln $dist_{ij}$		0.100		0.110		0.052		−0.041
		(0.024)		(0.022)		(0.021)		(0.043)
ln $popd_{ij}$	−0.011	−0.011	0.014	0.014	0.008	0.007	0.020	0.020
	(0.014)	(0.014)	(0.015)	(0.015)	(0.012)	(0.012)	(0.021)	(0.021)
Obs.	943	943	1,020	1,020	1,374	1,374	590	590
R^2	0.481	0.481	0.291	0.290	0.338	0.338	0.400	0.401
F-stat.	47.580	47.592	10.590	10.527	9.235	9.219	12.888	12.907

Note: The numbers in parentheses are White's heteroskedastic consistent standard errors. All regressions include product-specific dummy variables.

on the topographical conditions of the terrain. In Colombia, for instance, the (greater circle) distance between the cities of Bogotá and Cartagena is shorter than the existing one between Bogotá and Barranquilla. However, this result is no longer valid when distance is measured by moving along the road network, as, in order to go from Bogotá to Cartagena, one has to go to Barranquilla first. Because of these considerations, we opt for two alternative measures: the first one is the distance travelled through the road network between city pairs i and j (in kilometers); and the second one is the average time it takes to go from city i to city j (in minutes). Both measures are considered in logarithms and denoted $\ln dist_{ij}$ and $\ln durt_{ij}$, respectively.[3]

Examining the validity of the LOP, Esaka (2003) argues in favor of selecting similar cities, as deviations from equilibrium may occur because of variations in consumption preferences across cities and/or in population size. Thus, it appears conceivable to ask whether the degree of similarity of a city pair is also a factor that affects the speed at which relative prices adjust. To formally test for this, we consider relative population density as an additional regressor, measured by the absolute value of the difference of the logarithms of the population densities of cities i and j, which is $\ln popd_{ij} = |\ln pop_i - \ln pop_j|$; the source of these data is DANE.

The resulting regression model to examine the determinants of the (logarithm of the) half-lives of shocks affecting relative prices is therefore

$$\ln(hl_{ij,k}) = \beta_1 + \beta_2 \ln(popd_{ij}) + \beta_3 \ln(X_{ij}) + \text{Product Dummies} + \varepsilon_{ij,k}, \quad (2)$$

where X_{ij} is either $dist_{ij}$ or $durt_{ij}$, depending on the way the geographic separation of cities is measured, and $\varepsilon_{ij,k}$ is the error term in the regression. Notice that the model specified in Eq. (2) allows for product effects by means of the inclusion of product dummy variables. As customary, the estimate of the intercept term must be interpreted as the product against which comparisons are made, namely, potatoes, beef, medicines, and estimated rents for unprocessed foods, processed foods, other traded, and nontraded products, respectively.

Ordinary least squares estimation of the two variants of Eq. (2) yields the results summarized in Table 3. The estimated coefficient on $\ln dist_{ij}$ (or $\ln durt_{ij}$) is positive and statistically significant for all group categories, except for nontraded goods. This provides support for the view that the greater the geographical separation between cities, the longer it takes for their relative prices to adjust. Interestingly, this finding does not apply for the category of nontraded goods, where the estimated coefficient is not statistically different from zero; in other words, for goods that (for whatever reason) can only be consumed in the economy where they are produced, the distance to other markets is not a relevant factor affecting the speed at which their prices change, as would be expected. From a policy perspective, the results in Eq. (2) highlight the role that policy measures aimed at improving marketing infrastructure can play in securing (more) competitive markets all over the country.[4]

As to relative population density, the estimated coefficient on this variable does not turn out to be statistically different from zero in any of the regression models, which supports the view that the speed of adjustment of prices is not affected by the relative size of cities. Finally, in all the models, the coefficients associated with product-specific dummy variables (not reported for brevity) are jointly statistically different from zero, suggesting the need to account for product heterogeneity when modeling the speed of adjustment of relative prices.

[3] These variables are calculated using a personalized Java script that employs the Google Maps API v3 search engine. We are most grateful to Armando Galvis who kindly provided these data.

[4] In the international context, Goletti and Babu (1994) study the extent of maize markets integration in Malawi, and find that although market liberalization and price stabilization policies have served the purpose of enhancing market integration, the scope of these policies has been limited by developments on infrastructure and communications; see also the evidence for Mozambique in Cirera and Arndt (2008).

7. Concluding remarks

In this article, we use a disaggregated level CPI data for 153 goods to investigate the extent of market integration in Colombia. The empirical modeling exercise draws on a pairwise time-series approach, but in a way that also utilizes information from a cross-section dimension. The pairwise approach is based on the idea of examining the time-series properties of all possible city-pair price differentials for a given good. This is in contrast with the usual approach where attention is confined to the analysis of prices relative to a benchmark city, and therefore, the selection of this benchmark might affect the outcome. Our findings reveal strong support in favor of adjustment to the relative version of the LOP for unprocessed foods. By contrast, less support is found for processed foods, other traded and nontraded goods, which is perhaps not surprising for the latter category. We argue that the reason why a smaller number of price differentials are found to be stationary for processed foods and other traded products may have to do with unprocessed food products being more homogeneous, and also with the larger degree of market concentration that appears to characterize the manufacturing sector in Colombia.

A subsequent cross-section analysis of the speed of adjustment of city price differentials reveals an interesting spatial dimension in the way prices adjust to shocks in Colombia. Indeed, we uncover evidence that for unprocessed, processed, and traded goods, it takes more time for price shocks to disappear when cities are farther apart. This is regardless of whether one measures distance in kilometers through the road network, or as the time it takes to go from city to another. Distance does not appear to matter for nontraded goods. In a sense, one might think that the inclusion of the latter category serves the purposes of a control group. Finally, we find that the speed of adjustment of the prices is not affected by the relative size of cities (as measured by their population densities). In summary, our results support the role that policy measures aimed at improving marketing infrastructure can play at easing market integration across cities, and in facilitating prices to adjust quickly and homogeneously.

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Supporting Information

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Data Appendix